Jagadeeswara Chandra Bose K

Embedded Firmware & SoC Engineer → FPGA & Quantum Bit Control

krg.jcb@gmail.com · +91 97052 33003 · linkedin.com/in/jagadeeswara-k · jagadeeswara-k.github.io · Bangalore, IN

L0 · DEVICE	L1 · ANALOG	L2 · FPGA	L3 · FIRMWARE	L4 · CONTROL
Qubits	Power & signal	Control & timing	SoC bring-up	Estimation & feedback
TARGET	PROVEN	BUILDING	PROVEN	PROVEN

OBJECTIVE To build the **deterministic control layer** (FPGA timing, clean power and validated firmware) that turns fragile quantum hardware into **dependable applications**. I bring industry experience in **ARM64 SoC firmware bring-up and failure analysis** (Foxconn NPI) and **high-power battery/BMS validation** (Amara Raja R&D), plus **two filed patents** in real-time monitoring and control.

2	ARM64 SoC	600 V / 400 A	9.7 / 10
Patents filed, Indian Patent Office	NPI bring-up, Proto → EVT → DVT → PVT	Battery & power validation	M.Tech CGPA, Embedded & IoT

EXPERIENCE

System Firmware & OSD Failure Analysis Engineer

Jul 2025 – Present

Foxconn Hon Hai Technology India · Factory R&D, NPI

- **ARM64 SoC bring-up** (EFI, bootloader, kernel, OS/rootfs integration) across P0/P1/P2 → EVT → CRB → DVT → PVT.
- **Primary investigator** for kernel panics and system crashes; isolate hardware vs. firmware vs. configuration vs. software root cause.
- Component-, integration- and system-level validation with **Lua/Python automation** on test stations and jigs; smoke, regression and ERS command validation.
- Board-level debug with DSO, logic analyzer, JTAG/UART and USB debug; power-sequencing and boot analysis.
- Proposed and building a **Product Test Log Analyzer** to automate failure classification and speed up RCA. Author of SOPs and one-pagers for global teams.

Failure → Reproduce → Isolate → RCA → Fix → Regression → Production-ready

R&D Battery, BMS & Power Validation Engineer

Aug 2019 – Oct 2020

Amara Raja Energy & Mobility · R&D Laboratory, QA

- Validated Li-ion (3.2–3.8 V/cell, 20–200 Ah) and VRLA (12/14 V, 24–200 Ah) systems with CC-CV characterization.
- Ran a **600 V / 400 A battery cycler** for lifecycle, endurance and capacity tests; compliance, environmental and pre-customer qualification.
- Failure reproduction and RCA from V/I/T logs under ISO 17025 lab practice; technical reports for R&D and manufacturing.

EDUCATION

M.Tech · ECE, Embedded Systems & IoT
SRM University AP · CGPA 9.7/10
2023 – 2025 · FPGA, SoC, RTOS, AI/ML

B.Tech · Electrical & Electronics
SMGG, JNTUK · 71%
– 2022 · Power electronics

Diploma · Electrical & Electronics
SMGG, SBTET AP · 71%
2015 – 2018 · APGENCO 500 MW training

SSC (10th)
SRS Municipal High School, AP
2015

TOOLKIT

FPGA & SOC

Xilinx Zynq · Vivado · AXI/AMBA · ARM64 · Cortex-M · Tcl

FIRMWARE & CODE

C/Embedded C · C++ · Python · Lua · EFI/Bootloader · Linux debug · RTOS

ANALOG & POWER

DC-DC · GaN/MOSFET/IGBT · Gate drivers · LTspice · Altium · BMS

CONTROL & MODELLING

EKF · PI/SVPWM · MATLAB/Simulink · Digital twin

PATENTS

PATENT FILED · IPO 202641069435

Digital Twin-Based Multi-Bank Battery Monitoring & Fault Prediction

5-layer sensor → edge → AI → cloud → dashboard twin for 150 cells. Hybrid estimators (adaptive EKF, Li-Mamba, ensemble) run against live hardware.

0.65% SOC error · **96.2%** early fault detection · **<2 s** alerts · 22 health parameters

Solves: battery packs that fail without warning, with early fault alerts and accurate SOC for EV and storage systems.

PATENT FILED · IPO 202541009491

System & Method for Managing EV Charging Infrastructure through IoT

5-layer IoT architecture (MQTT, AWS) with a source-priority decision algorithm across 30 swappable batteries.

88.6% energy saving · **88.5%** cost saving · real-time remote control

Solves: costly EV charging, by choosing the best energy source in real time and managing battery swapping remotely.

RESEARCH & PROJECTS

SEED GRANT · PAPER UNDER REVIEW

IoT-Enabled Hybrid Energy Management System for EV Charging (hardware prototype)

Dual ESP32/PIC controller with 5-mode PCC logic, ACS712/DHT11 sensing; solved edge-layer timing jitter. SRM AP SEED grant SRMAP/URG/SEED/2024-25/038, part of DST-SERB / ANRF research with Amara Raja.

<200 ms switching · **86.2%** renewable utilization · reverse grid export

Solves: reliable renewable-powered EV charging with fast solar / battery / grid switching.

IIT PALAKKAD · SUMMER RESEARCH FELLOW · 2024

Power Electronic Circuits: Numerical Simulations

Designed and simulated 7 DC-DC topologies (buck, boost, buck-boost, forward, flyback, push-pull, synchronous buck); custom GaN MOSFET and gate-driver libraries in LTspice; 2D/3D Altium PCB for a sub-zero battery pre-heater. Mentor: Dr. N. B. Y. Gorla.

7 topologies · **30%** faster design iterations

Solves: slow converter design cycles, using formula sheets plus SPICE validation.

MODELLING · MATLAB/SIMULINK

Modelling & Simulation of an HEV Electric Powertrain

PMSM with 3-phase 2-level VSI and SVPWM; closed-loop PI speed/current control validated over the IM240 cycle (torque, speed, SOC, switching).

Solves: checking a hybrid powertrain before any hardware is built.

EMERTXE · 2022-23

Mongoose OS on ESP32 for AWS IoT

OTA firmware updates and secure device management; MQTT telemetry into AWS IoT Core, Analytics and QuickSight pipeline.

FPGA → QUBIT PATH

NOW	Qiskit, circuit QED, RF/IQ basics, RTL on Zynq
NEXT	Zynq pulse sequencer, FPGA IQ demod, low-noise bias supply
LATER	Scalable qubit control electronics & feedback

LAB & INTERFACES

LAB

DSO · Logic analyzer · JTAG/UART · Battery cyclers · ISO 17025

INTERFACES

SPI · I²C · UART · CAN · PCIe · USB · MQTT

CERTIFICATIONS

Embedded Systems & IoT
Emertxe Information Technologies, Bangalore
2022 – 2023 · ARM, RTOS, MQTT/CoAP, AWS

AI / ML Program
IIT Patna (Masai School)
Ongoing · ML, LLMs, RAG, Agentic AI

A2I · ACADEMIC TO INDUSTRY

I host an **Engineering School platform** that helps academic students move into industry through hands-on projects, engineering practice and mentoring from people in the field.

Design Thinking · Project Management · Product Development · Prototyping · Cross-functional collaboration · Technical documentation · 7 QC tools

LINKS

Portfolio · jagadeeswara-k.github.io
GitHub · Jagadeeswara-k
Medium · @krg.jcb
Kaggle · jagadeeswarak

I declare that the information above is accurate to the best of my knowledge.